

A

B

C

D

E

4

3

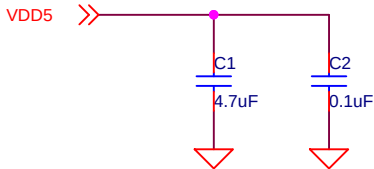
2

1

RTL8152B 24 Pin Reference Schematic V103

PAGE	INDEX
1.	Title Page
2.	POWER
3.	CONTROLLER
4.	Reserved Components
5.	Bill Of Materials
6.	Change Notes

5V



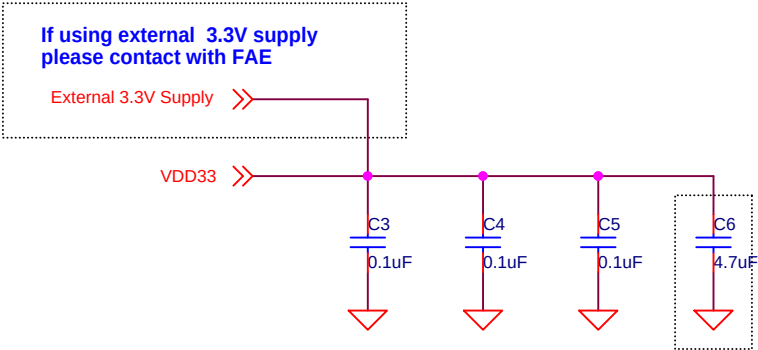
Place C1 and C2 close to VDD5 pin-- 11

Note

For RTL8152B

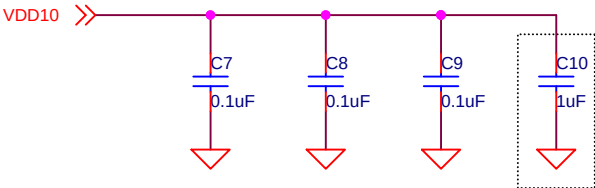
- 5V power input pin-- 11
- 3.3V LDO output pin-- 10
- 3.3V power input pin-- 1, 10, 13
- 1.05V LDO output pin-- 23
- 1.05V power input pin-- 9, 16, 23
- 1.05V UPS power pin-- 12

3.3V

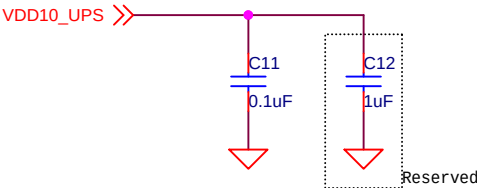


Place C3 to C5 close to each VDD33 pin-- 1, 10, 13
Place C6 close to VDD33 pin-- 10 (reserved)

1.05V



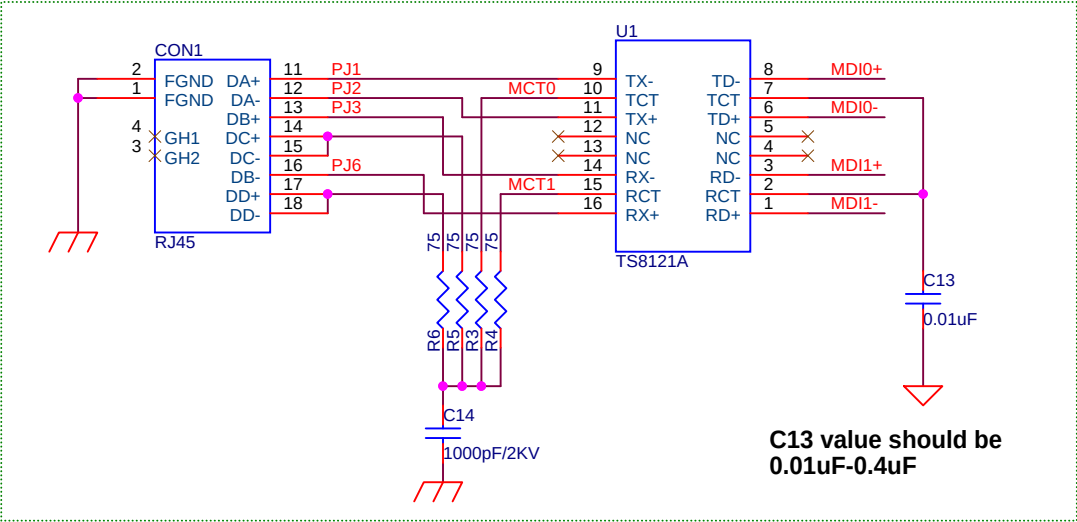
Place C7 to C9 close to each VDD10 pin-- 9, 16, 23
Place C10 close to VDD10 pin-- 9



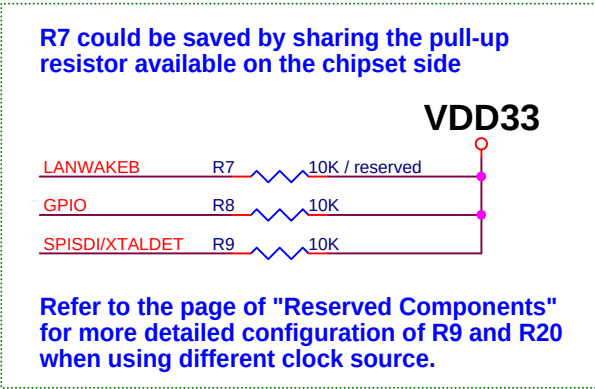
Place C11 close to VDD10_UPS pin-- 12
Place C12 close to VDD10_UPS pin-- 12 (reserved)

Realtek Semiconductor Corp.			
No.2, Industry E. Rd. IX, Science-Based Industrial Park, Hsinchu, Taiwan 300			
Title RTL8152B 24 Pin Reference Schematic : Power			
Size A4	Document Number RTL8152B 24 Pin Reference Schematic		Rev
Date:	Wednesday, January 30, 2013	Sheet	2 of 6

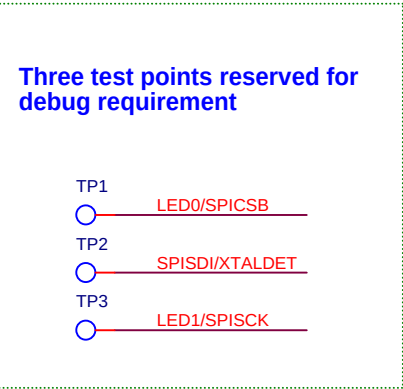
RJ-45 Connector + Transformer



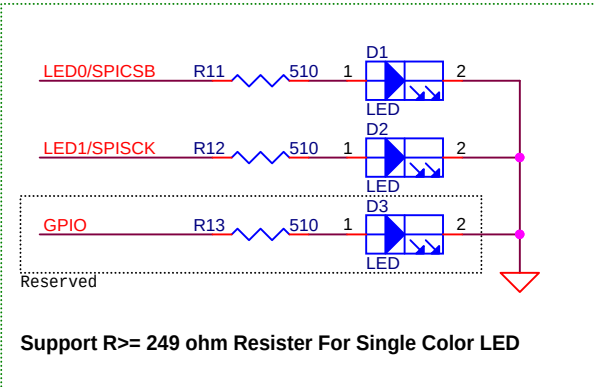
Pull-up Resistor



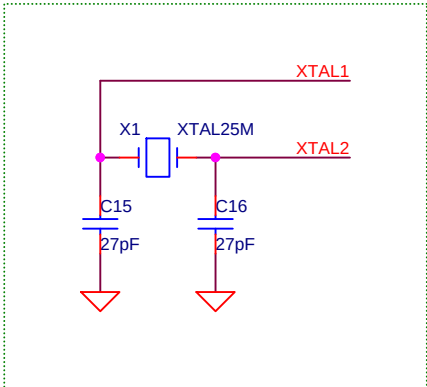
Test Point



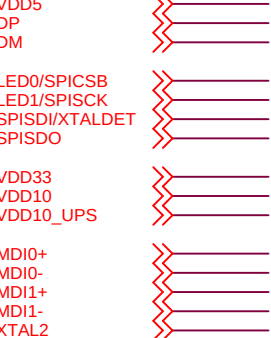
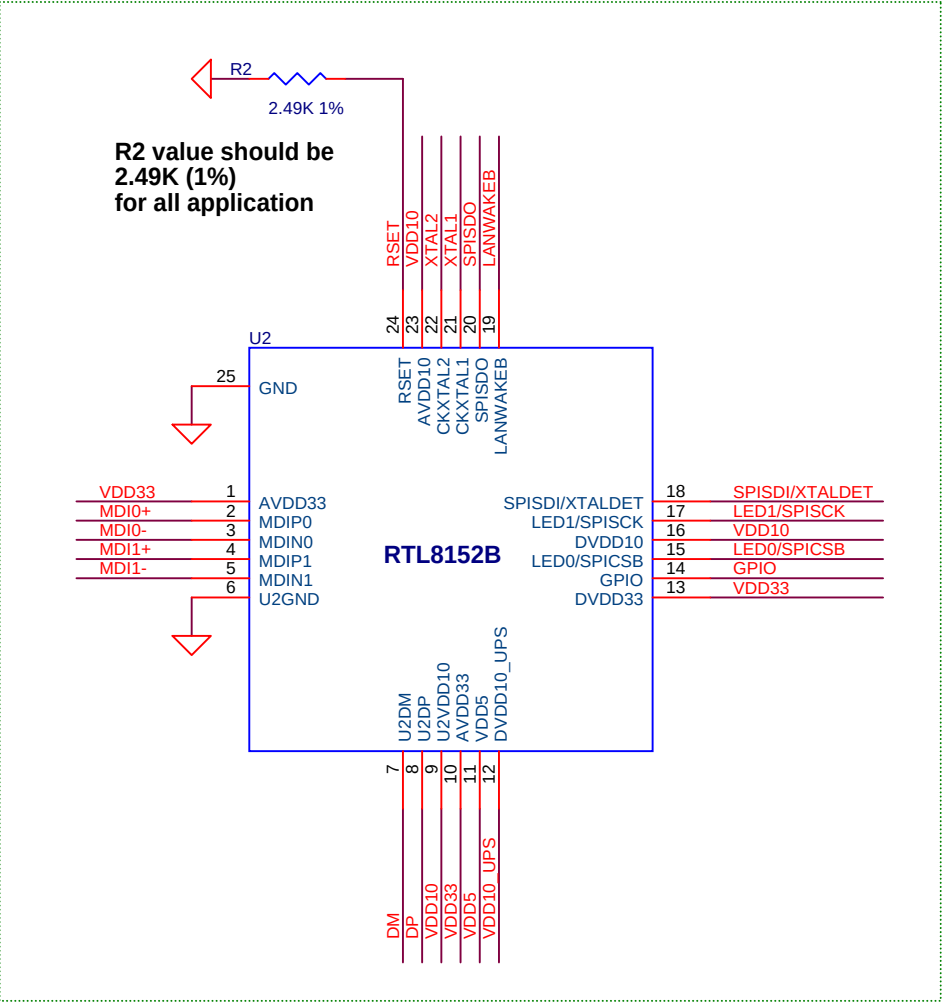
LED (High Active)



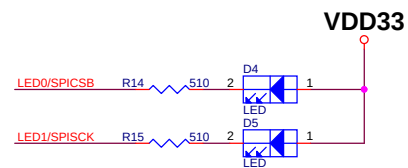
XTAL



Controller

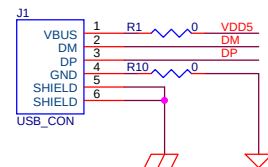


LED (Low Active)



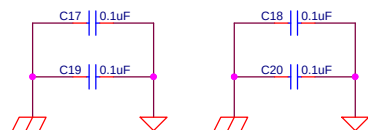
Support R>= 249 ohm Resister For Single Color LED

USB Connector



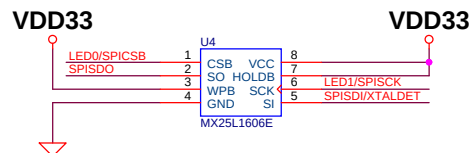
R1, R10 are reserved for filtering power noise.
You can just make a path of short to replace the use of R1/R10 if not needed.

Reserved for Noise Isolation

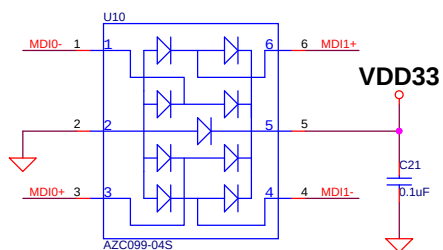


C17 to C20 Reserved for Noise Isolation between Chassis GND and System GND (optional)

SPI Flash

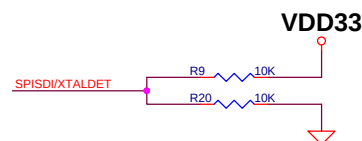


Reserved for TVS Components



Note:
1. U10 and C21 are reserved for line-to-line Surge and Cable ESD on the secondary side (optional)
2. Gas Tube is for line-to-ground Surge on the primary side (optional)

Reserved for Using External Clock Source



R9 Reserved for Using 25MHz Clock Source
R20 Reserved for Using 48MHz Clock Source

Pin18- XTALDET is Clock Detect Pin.
Keep R9 and Remove R20 when 25MHz clock is used.
Keep R20 and Remove R9 when 48MHz clock is used.

R21 Reserved for Using External Clock Source (optional)

Pin22- XTAL2 is External Clock Input Pin
Pin21- XTAL1 must be floating when using external clock source



Realtek Semiconductor Corp.		
No.2, Industry E. Rd. IX, Science-Based Industrial Park, Hsinchu, Taiwan 300		
Title		
RTL8152B 24 Pin Reference Schematic: Reserved Components		
Size	Document Number	Rev
A3	RTL8152B 24 Pin Reference Schematic	
Date:	Wednesday, January 30, 2013	Sheet 4 of 6

1. R-BOM for LAN on Motherboard

Bill Of Materials January 30, 2013

Item	Quantity	Reference	Part
1.	1	CON1	RJ45
2.	1	C1	4.7uF
3.	8	C2,C3,C4,C5,C7,C8,C9,C11	0.1uF
4.	1	C10	1uF
5.	1	C13	0.01uF
6.	1	C14	1000pF/2KV
7.	2	C15,C16	27pF
8.	2	D1,D2	LED
9.	1	J1	USB2.0 Connector
10.	1	R2	2.49K 1%
11.	4	R3,R4,R5,R6	75
12.	2	R8,R9	10K
13.	2	R11,R12	510
14.	1	U1	Transformer
15.	1	U2	RTL8152B
16.	1	X1	XTAL25M

2. R-BOM for USB dongle

Bill Of Materials January 30, 2013

Item	Quantity	Reference	Part
1.	1	CON1	RJ45
2.	1	C1	4.7uF
3.	8	C2,C3,C4,C5,C7,C8,C9,C11	0.1uF
4.	1	C10	1uF
5.	1	C13	0.01uF
6.	1	C14	1000pF/2KV
7.	2	C15,C16	27pF
8.	2	D1,D2	LED
9.	1	J1	USB2.0 Connector
10.	1	R2	2.49K 1%
11.	4	R3,R4,R5,R6	75
12.	2	R8,R9	10K
13.	2	R11,R12	510
14.	1	U1	Transformer
15.	1	U2	RTL8152B
16.	1	U4	SPI Flash
17.	1	X1	XTAL25M

Realtek Semiconductor Corp.

No.2, Industry E. Rd. IX, Science-Based Industrial Park, Hsinchu, Taiwan 300

Title
RTL8152B 24 Pin Reference Schematic : Bill Of Materials

Size A4	Document Number RTL8152B 24 Pin Reference Schematic	Rev
------------	--	-----

Date: Wednesday, January 30, 2013	Sheet 5 of 6
-----------------------------------	--------------

Change Notice from 1.00 to 1.01

1. Add R5, R6 for better EMI suppression. (page 3)
2. Modify R-BOM list. (page 5)

Change Notice from 1.01 to 1.02

1. Add LED configurations to high active for better system ESD protection. (Page 3)

Change Notice from 1.02 to 1.03

1. Add C6 note for surge improvement. (Page 2)
2. Add C10 for power stability. (Page 2, Page 5)

Realtek Semiconductor Corp.		
No.2, Industry E. Rd. IX, Science-Based Industrial Park, Hsinchu, Taiwan 300		
Title		
RTL8152B 24 Pin Reference Schematic : Change Notes		
Size	Document Number	Rev
A4	RTL8152B 24 Pin Reference Schematic	
Date:	Wednesday, January 30, 2013	Sheet 6 of 6